



// INSTALLATION STEPS

Ubuntu 24.04 + ROS 2 Jazzy + Gazebo Harmonic

Everything to install before the workshop, from a fresh OS to a working simulation, with a check at the end to confirm it all runs.

```
user@robot:~$ ros2 run demo_nodes_cpp talker
```

OPERATING SYSTEM

Ubuntu 24.04

Noble Numbat · native install

MIDDLEWARE

ROS 2 Jazzy

Jazzy Jalisco · LTS

SIMULATOR

Gazebo Harmonic

with ros_gz bridge

00

Before you begin

IMPORTANT

We strongly recommend that workshop attendees do **not** use a Virtual Machine for this setup. Install Ubuntu natively, either as a dual boot or on a dedicated drive.

- ☐ Install **Ubuntu 24.04 LTS** (Noble Numbat).
- ☐ **ROS 2 Jazzy Jalisco** is built for Ubuntu 24.04 (Noble). Other Ubuntu versions will not work with these steps.
- ☐ Back up your data before you partition a disk for dual boot.

01

Install Ubuntu 24.04

Download the desktop image, write it to a USB drive, and follow the official installer walkthrough.

DOWNLOAD Ubuntu 24.04 Desktop
<https://ubuntu.com/download/desktop>

GUIDE Installation guide
<https://ubuntu.com/tutorials/install-ubuntu-desktop#2-download-an-ubuntu-image>

VIDEO Installation walkthrough
<https://www.youtube.com/watch?v=aIFosqQ1ang>

02

Install ROS 2 Jazzy

Follow the official Debian-package guide. It adds the ROS apt repository and installs `ros-jazzy-desktop`.

DOCS ROS 2 Jazzy · Ubuntu (deb)
<https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html>

VIDEO ROS 2 Jazzy walkthrough
<https://www.youtube.com/watch?v=08o46x5SfJM>



03

Environment setup

Once ROS 2 Jazzy is installed, set up your environment by sourcing its setup file.

SOURCE ROS IN THE CURRENT TERMINAL

```
● ● ● bash  
  
$ source /opt/ros/jazzy/setup.bash
```

LOAD IT AUTOMATICALLY IN EVERY NEW TERMINAL

```
● ● ● bash  
  
$ echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc  
$ source ~/.bashrc
```

NOTE

Running `source ~/.bashrc` reloads the updated file. Closing all terminals and opening a new one does the same thing.

04

Install Gazebo Harmonic

Harmonic is the Gazebo release paired with ROS 2 Jazzy.

DOCS Harmonic • Ubuntu install

https://gazebosim.org/docs/harmonic/install_ubuntu/

VIDEO Gazebo walkthrough

<https://www.youtube.com/watch?v=dY8JxldcuqA>

DOCS Getting started with Gazebo

<https://gazebosim.org/docs/harmonic/getstarted/>

05 ROS ↔ Gazebo integration

Install the pairing packages that let ROS 2 nodes talk to Gazebo simulations. Reference: gazebosim.org/docs/harmonic/ros_installation

DEFAULT ROS-GAZEBO PAIRING

```
● ● ● bash
$ sudo apt-get install ros-jazzy-ros-gz
```

BRIDGE FOR TOPIC COMMUNICATION BETWEEN ROS 2 AND GAZEBO

```
● ● ● bash
$ sudo apt-get install ros-jazzy-ros-gz-bridge
```

ROS2_CONTROL PLUGIN FOR LOADING AND RUNNING CONTROLLERS IN GAZEBO

```
● ● ● bash
$ sudo apt-get install ros-jazzy-gz-ros2-control ros-jazzy-gz-ros2-control-demos
```

06 Key packages

The packages used throughout the workshop for control, robot description, navigation and mapping.

<code>ros2_control</code> • <code>ros2_controllers</code>	Controller manager and controllers
<code>xacro</code>	Macros for robot description files
<code>navigation2</code> • <code>nav2_bringup</code>	Navigation2 (Nav2) and its bringup launch files
<code>slam_toolbox</code>	SLAM, for building maps
<code>twist_mux</code> • <code>twist_stamper</code>	Multiplexing and stamping velocity commands

```
● ● ● bash

# ros2_control (controller manager, controllers)
$ sudo apt install ros-jazzy-ros2-control ros-jazzy-ros2-controllers
# xacro (robot description macros)
$ sudo apt install ros-jazzy-xacro
# Navigation2 (Nav2) + bringup
$ sudo apt install ros-jazzy-navigation2 ros-jazzy-nav2-bringup
# SLAM
$ sudo apt install ros-jazzy-slam-toolbox
# Twist mux and Twist stamper
$ sudo apt install ros-jazzy-twist-mux
$ sudo apt install ros-jazzy-twist-stamper
```

07 Verify the installation

After finishing all the steps above, run these in **new** terminals to confirm that ROS and Gazebo both work.

TERMINAL A • C++ TALKER

```
terminal A

$ ros2 run demo_nodes_cpp talker
[INFO] [talker]: Publishing: 'Hello World: 1'
```

TERMINAL B • PYTHON LISTENER

```
terminal B

$ ros2 run demo_nodes_py listener
[INFO] [listener]: I heard: [Hello World: 1]
```

The talker should report that it is publishing messages, and the listener should report that it heard them. This confirms that both the C++ and Python APIs work.

GAZEBO HARMONIC

```
bash

$ gz sim shapes.sdf
```

This starts the Gazebo server and GUI with a sample world containing three simple shapes.



Quick fixes

SYMPTOM	TRY THIS
ros2: command not found	The terminal has not sourced ROS. Run <code>source /opt/ros/jazzy/setup.bash</code> or redo step 03.
Unable to locate package ros-jazzy-...	The ROS apt repository was not added, or you are not on Ubuntu 24.04. Recheck step 02 and run <code>sudo apt update</code> .
Listener hears nothing	Make sure both terminals are sourced and have the same <code>ROS_DOMAIN_ID</code> (check with <code>printenv ROS_DOMAIN_ID</code>).



You're all set.

If all three checks pass, ROS 2 Jazzy and Gazebo Harmonic are installed and working on your system.